

2024



Flywire Improves Speed to Deployment by 60% Using Amazon ECS with AWS Fargate

Learn how payment software company Flywire modernized its cloud environment using Amazon ECS with AWS Fargate.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

60%

faster startup times for development pipelines



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time saved on manual security patches

70%

compute cost savings achieved



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Overview

Global payments enablement and software company [Flywire](#) has cultivated a global customer base across the healthcare, education, business, and travel sectors through a combination of strong organic growth and strategic acquisitions. As the company expanded into new markets and verticals, it wanted to rearchitect its cloud environment to automatically scale with demand and to optimize compute costs.

To free its teams from having to scale clusters and manage servers, Flywire containerized its workloads and adopted serverless solutions on Amazon Web Services (AWS). Since modernizing in the cloud, Flywire has scaled from running 3 to 150 services on AWS while saving up to 70 percent on compute costs.



Opportunity | Reducing Heavy Lifting for Flywire's Engineering Teams using Amazon ECS with AWS Fargate

Founded in 2011, Flywire originated from one student's vision to simplify his tuition payments while studying abroad. Since then, the company's mission has evolved into delivering some of the world's most important and complex payments to its clients across multiple industries, underpinned by the company's proprietary global payment network, next-gen payments platform, and industry-specific software. The company has experienced year-over-year growth and serves more than 3,700 global clients with diverse payment methods in more than 140 currencies across 240 countries and territories around the world.

As Flywire expanded, it migrated to AWS, seeking increased scalability, security, and efficiency. Following its initial migration, Flywire wanted to free its teams from self-managing its AWS resources. "We aim to make our products more resilient and to improve the lives of our engineers," says José Luis Salas, head of site reliability engineering at Flywire. "Instead of patching servers, we wanted to focus on supporting the business."

In 2018, Flywire migrated its workloads to [Amazon Elastic Container Service](#) (Amazon ECS), which runs highly secure, reliable, and scalable containers. To further boost engineering productivity, Flywire decided to run Amazon ECS with [AWS Fargate](#), a serverless, pay-as-you-go compute engine that gives Flywire's teams the ability to build applications without managing servers. By running its containers on AWS Fargate, Flywire's teams up to 40 percent of their time on manual security patches. "Amazon ECS with AWS Fargate has been the



perfect combination for us,” says Salas. “AWS Fargate integrates well with other AWS services with almost zero maintenance.”

Solution | Improving Security on AWS and Achieving 60% Faster Startup Times Using SOCI with AWS Fargate

As Flywire rearchitected its environment, cloud security was its top priority. Flywire must comply with SOC 2, HIPAA, ISO, and other industry standards. Flywire bolstered its security posture by adopting [AWS Security Hub](#), which automates AWS security checks and centralizes security alerts, and [Amazon GuardDuty](#), which protects Flywire’s AWS accounts with intelligent threat detection. As the company spun up more than 700 clusters on Amazon ECS, its site reliability engineering team benefited from fully managed services. “There’s zero additional maintenance for that team,” says Salas. Additionally, the team achieved 70 percent cost savings by running its development environment on [AWS Fargate Spot](#), which can run fault-tolerant Amazon ECS tasks at up to a 70 percent discount off Amazon EC2 On-Demand Pricing.

Throughout the project, Flywire worked alongside a complete account team within AWS as well as [AWS Enterprise Support](#), which provides a suite of resources to help organizations scale, optimize costs, and accelerate value from their cloud investment. Together, the teams focused on architecting an environment that would support operational excellence and efficiency. To avoid misconfigurations, Flywire has set up development, testing, and production environments using AWS Fargate. “The ability to automatically scale on Amazon ECS has meant that we can rely on a cloud security team of three people,” says Salas. By designing its architecture with three separate environments, Flywire’s engineers can rapidly test out new ideas by spinning up resources under a self-service model that has preset controls and guardrails.

After experiencing the benefits of going serverless, Flywire enrolled in the private preview for the [Seekable OCI \(SOCI\)](#) feature, a technology open sourced by AWS that helps containers launch faster by lazily loading the container image. “We want to be early adopters of AWS technology,” says Salas. “By participating in beta tests and private previews, we’re able to contribute feedback during the development of features and solutions.”

With this feature, the Flywire team saw an immediate improvement in startup times, helping and release features quickly. “Using SOCI with AWS Fargate was amazing and saved us a lot of time.” “Our pipelines run 60 percent faster than before.”



Outcome | Expanding to New Markets Using a Serverless Stack

Flywire's modernized, serverless stack, was a benefit when the company scaled its operations into Canada and Australia. The company will continue to identify opportunities for business growth using the latest cloud technology. "AWS has been a key player in our growth. It has a managed service for almost any business need we have," says Salas. "We have the tools to test new ideas with little effort. One of the biggest benefits of working on AWS is our ability to innovate."

About Flywire

Flywire is a global payments enablement and software company that combines its proprietary global payments network, next-gen payments platform, and vertical-specific software to deliver the most important and complex payments to global clients.

AWS Services Used

Amazon ECS

Amazon Elastic Container Service (Amazon ECS) is a fully managed container orchestration service that simplifies your deployment, management, and scaling of containerized applications.

[Learn more »](#)

AWS Fargate

AWS Fargate is a serverless, pay-as-you-go compute engine that lets you focus on building applications without managing servers. AWS Fargate is compatible with both Amazon Elastic Container Service (Amazon ECS) and Amazon Elastic Kubernetes Service (Amazon EKS).

[Learn more »](#)

AWS Security Hub

AWS Security Hub is a cloud security posture management (CSPM) service that performs security practice checks, aggregates alerts, and enables automated remediation.



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Amazon GuardDuty

Amazon GuardDuty is a threat detection service that continuously monitors your AWS accounts and workloads for malicious activity and delivers detailed security findings for visibility and remediation.

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